

Robust RPC Bundle Adjustment for Multi-Date Satellite Imagery with Season-Invariant Correspondences

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Motivation & Contributions

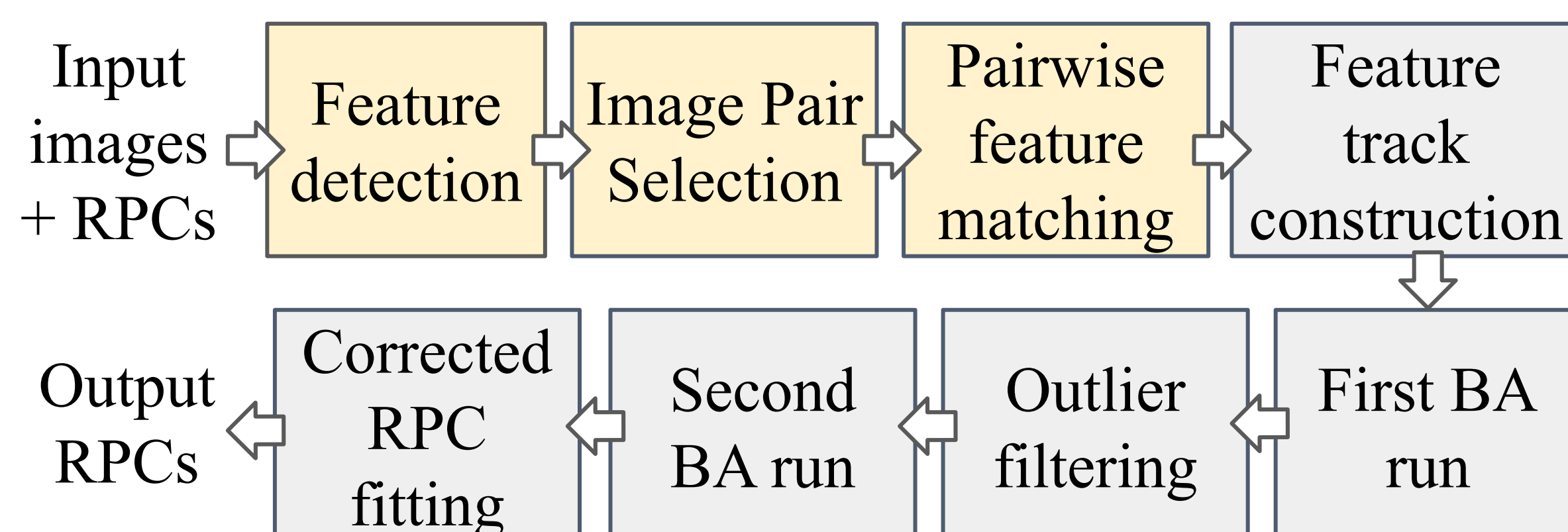
The performance of existing RPC bundle adjustment tools degrades for diachronic image collections affected by seasonal, illumination, and land-cover changes.

We propose:

1. A systematic benchmark and open-source integration of learned local matchers and image descriptors into an RPC BA pipeline.
2. A novel image-pair selection strategy based on global visual similarity.

RPC Bundle Adjustment

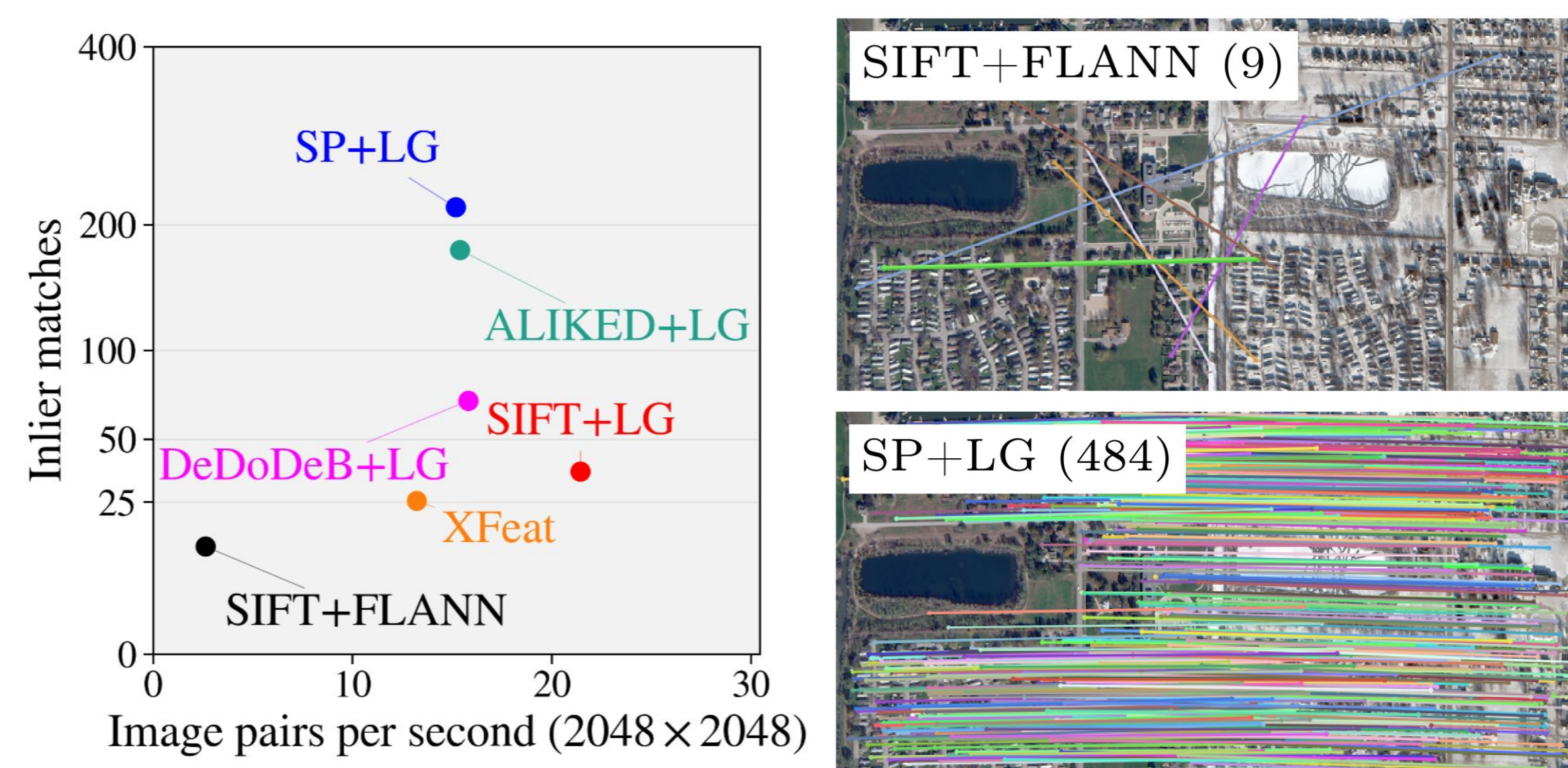
Pipeline overview:



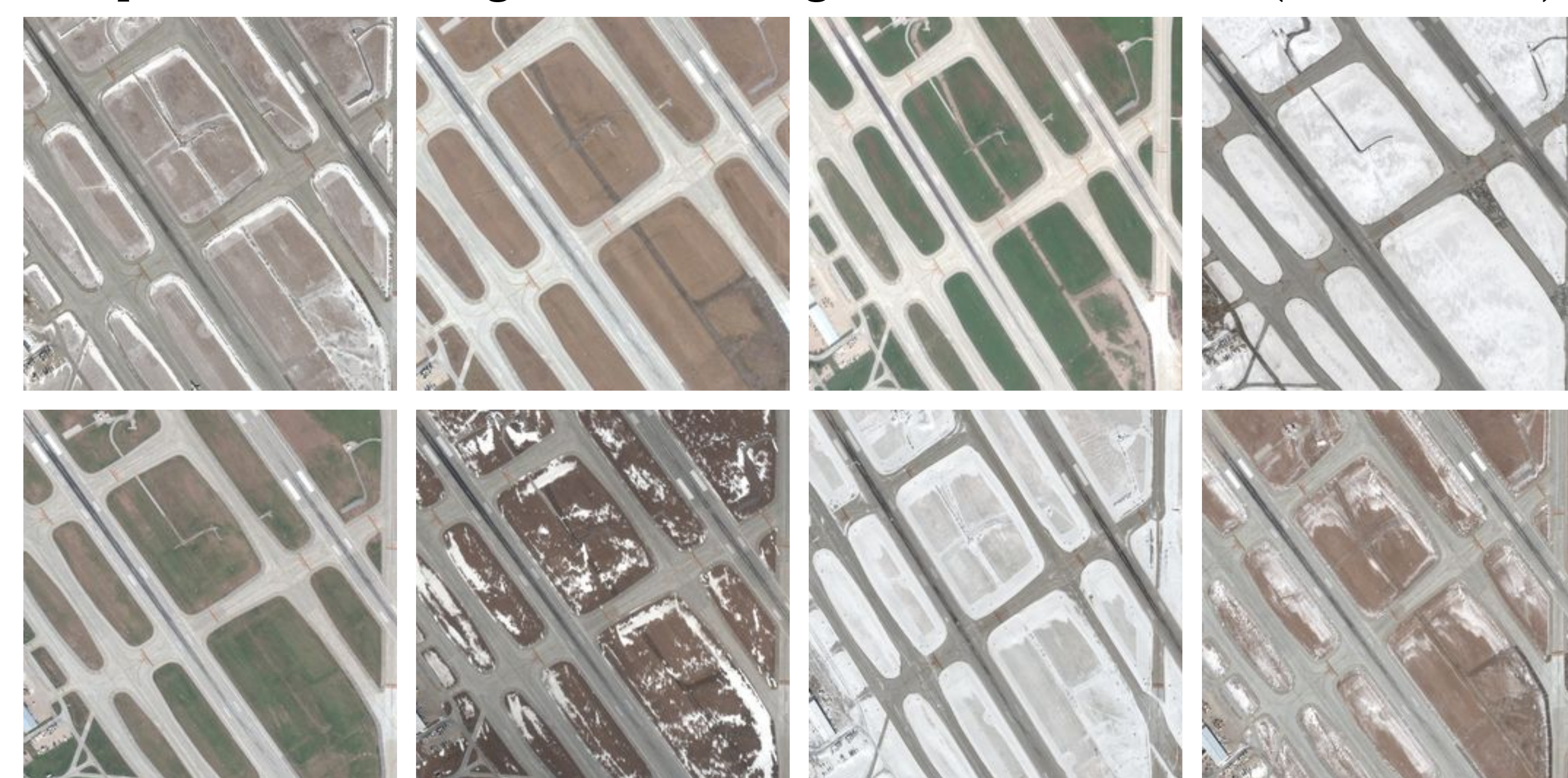
Yellow steps show benchmarked components and the proposed image-pair selection.

Season-Invariant Feature Matching

Benchmarking detector-based matchers reveals that SuperPoint+LightGlue feature matching yields the largest number of local matches across diachronic image pairs.



Input satellite images with strong seasonal variation (~40 in total)



SAT-BA v1 (~800 pairs) Ours (~200 pairs)

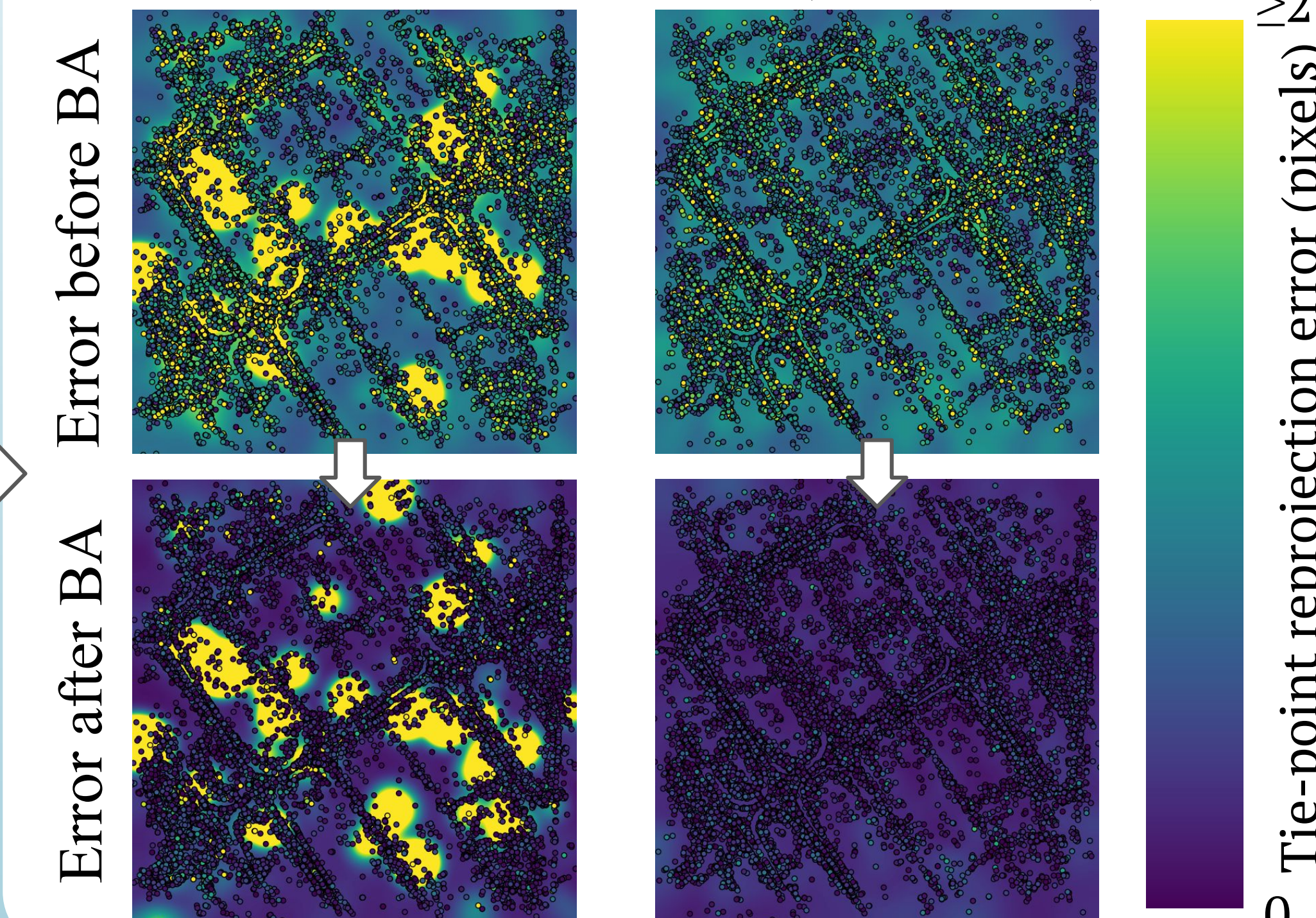


Image Pair Similarity Estimation

After benchmarking different global image descriptors, DINOv2 cosine similarity is selected to score image pairs. DINOv2 shows the strongest correlation with the number of verified local matches.

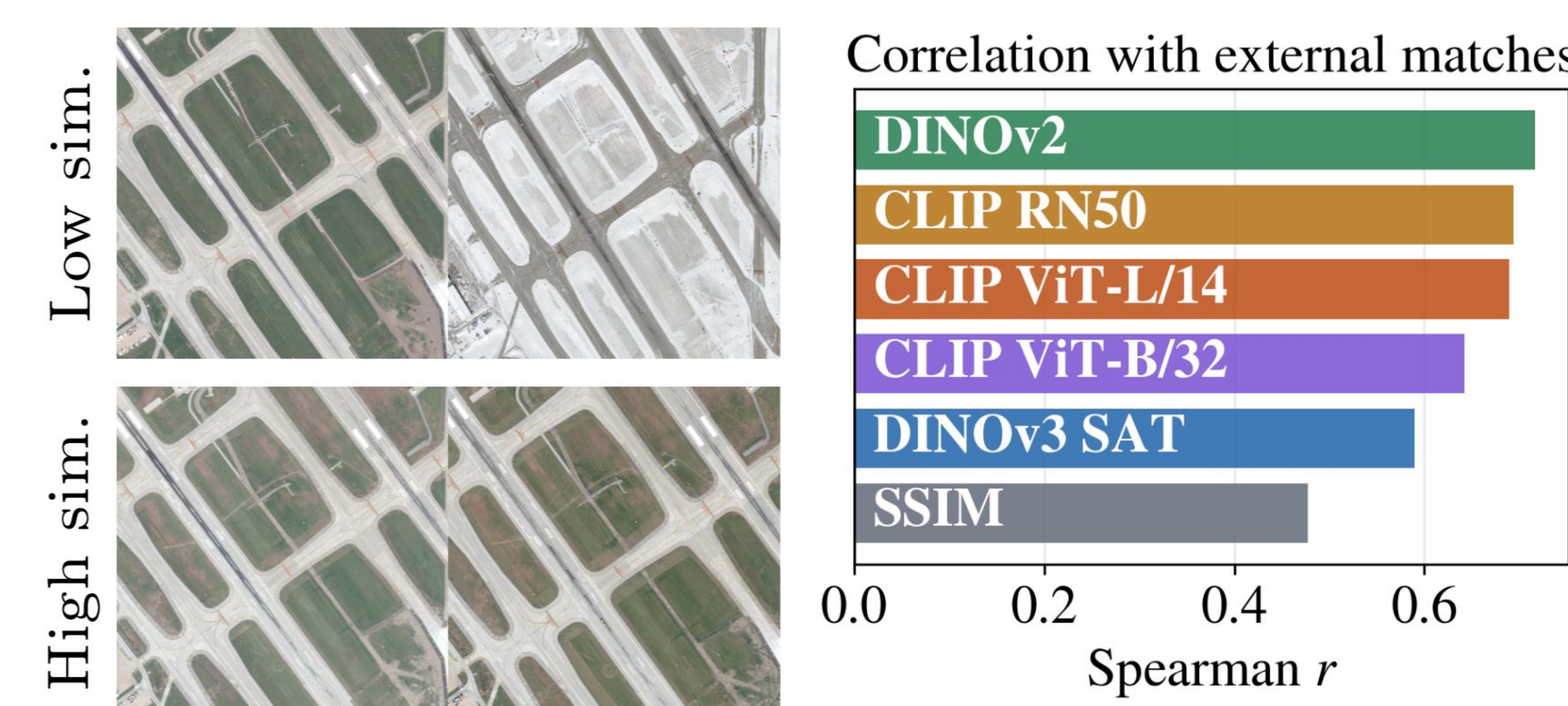
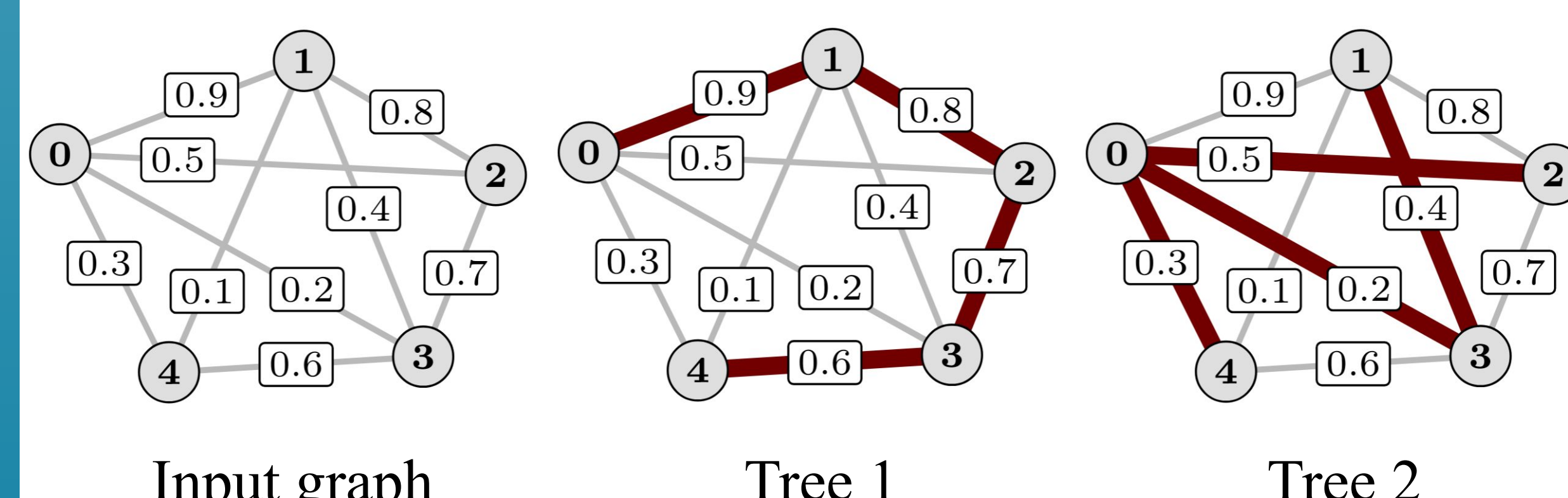


Image Pair Selection via Similarity

Pair selection is formulated as a weighted graph problem. Let $G = (V, E)$, where V is the set of images (nodes) and E is the set of candidate pairs (edges), each weighted by a similarity score. The algorithm iteratively selects K edge-disjoint maximum spanning trees, prioritizing the most similar pairs.



Datasets

- DFC2019 *Track3* multi-view WV-3 RGB image crops of 2048x2048 pixels. [1]
- **30 multi-date diachronic collections** over Omaha (OMA) areas of interest, each containing 39-42 image crops.
- **11 multi-date synchronic collections** over Jacksonville (JAX) areas of interest, each containing 24 image crops.

Baseline Methods

- Ames Stereo Pipeline (ASP). [2]
- Sat-BundleAdjust v1 (SAT-BAv1). [3]

Both baselines implement RPC BA using handcrafted feature matching to establish image correspondences and export corrected camera models in RPC format.

Evaluation Metrics

Corrected RPCs are evaluated using external pseudo-GT feature tracks excluded from all bundle adjustment optimizations.

- Held-out reprojection errors.
- Pairwise 3D consistency errors.

Results



Try it yourself!

<https://centreborelli.github.io/sat-bundleadjust/v2>

* GPU support needed.

Method	Matcher	# Pairs	# Obs/Cam	Err. [px]	Time [min:s]
ASP	SIFT + FLANN	~807	~289	1.458	18:06
SAT-BAv1	SIFT + FLANN	~807	~904	2.353	05:45
Ours	SP + LG	~198	~1670	0.422	02:22

Internal stats (Omaha collections). Fastest in red.

Method	Held-out reprojection errors [px]			Pairwise 3D consistency errors [m]		
	Mean	Median	P90	Height MAD	Median	P90
Raw RPCs	1.94	1.75	3.43	0.59	0.78	1.78
ASP	1.53	1.28	2.92	0.47	0.62	1.44
SAT-BAv1	11.08	7.14	27.50	2.52	3.94	8.86
Ours	1.34	1.11	2.62	0.39	0.55	1.23

Evaluation metrics (Omaha). Top RPC accuracy in red.

Experiments on the Omaha collections show that our contributions improve output RPC accuracy on the target diachronic WV-3 imagery while reducing matching time.

Ablation studies on the Jacksonville collections further show that similarity-based pair selection reduces matching time without degrading performance in the synchronic setting.

References

- [1] Bosch et al. "Semantic stereo for incidental satellite images.", *WACV*, 2019.
- [2] Beyer et al. "The Ames Stereo Pipeline: NASA's open source software for deriving and processing terrain data." *Earth and Space Science*, 2018.
- [3] Marí et al. "A generic bundle adjustment methodology for indirect RPC model refinement of satellite imagery." *Image Processing On Line*, 2021.